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Electricity consumption and economic growth in South Africa: A trivariate causality test

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ABSTRACT

In this paper we examine the causal relationship between electricity consumption and economic growth in South Africa. We incorporate the employment rate as an intermittent variable in the bivariate model between electricity consumption and economic growth—thereby creating a simple trivariate causality framework. Our empirical results show that there is a distinct bidirectional causality between electricity consumption and economic growth in South Africa. In addition, the results show that employment in South Africa Granger-causes economic growth. The results apply irrespective of whether the causality is estimated in the short-run or in the long-run formulation. The study, therefore, recommends that policies geared towards the expansion of the electricity infrastructure should be intensified in South Africa in order to cope with the increasing demand exerted by the country's strong economic growth and rapid industrialisation programme. This will certainly enable the country to avoid unprecedented power outages similar to those experienced in the country in mid-January 2008.

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1. Introduction

The debate regarding the causal relationship between electricity consumption and economic growth has attracted increasing attention in recent years. The thrust of this debate has been whether electricity consumption Granger-causes economic growth or whether it is economic growth that Granger-causes electricity consumption. From a policy standpoint, the causal relationship between electricity consumption and economic growth has important implications (see Asafu-Adjaye, 2000; Ghosh, 2002; Paul and Bhattachrya, 2004; Narayan and Smyth, 2005; Narayan and Singh, 2007). A finding in favour of a positive unidirectional causality running from GDP to electricity consumption may imply that a country is not entirely dependent on electricity for its economic growth, and that energy conservation policies may be implemented with little or no adverse effects on economic growth. However, a unidirectional causality running from electricity consumption to economic growth implies that economic growth is dependent on electricity consumption and that a decrease in electricity consumption will inevitably lead to a decrease in real income, which may have a negative effect on employment and job creation (see also Narayan and Singh, 2007:1142). The latter outcome may have far-reaching implications for a country like South Africa, which is currently targeting a sustained GDP growth rate of 6% by the year 2010. The finding of no causality in either direction, i.e. the so-called 'neutrality hypothesis', on the other hand, implies that energy conservation policies do not have a significant effect on economic growth (see Asafu-Adjaye, 2000; Paul and Bhattachrya, 2004).

Although the debate regarding the causal relationship between energy and economic growth has generated much literature in both developed and developing countries, the majority of the previous studies concentrated mainly on Asia and Latin America, affording sub-Saharan African countries either very little or no coverage at all. Even where such studies have been undertaken, the empirical findings on the direction of causality between electricity consumption and economic growth have been largely inconclusive. Previous studies on this subject, however, suffer from two major limitations. Firstly, some of the studies have over-relied on the cross-sectional data which cannot satisfactorily address the country-specific issues. Secondly, some studies have used a bivariate framework, and may, therefore, suffer from the omission of variables bias. In other words, the introduction of a third variable affecting both electricity consumption and economic growth in the bivariate framework may not only alter the direction of causality between the two variables, but also the magnitude of the estimate. It is against this backdrop that the current study attempts to examine the intertemporal causal relationship between electricity consumption and economic growth in South Africa by incorporating the rate of unemployment as an intermittent

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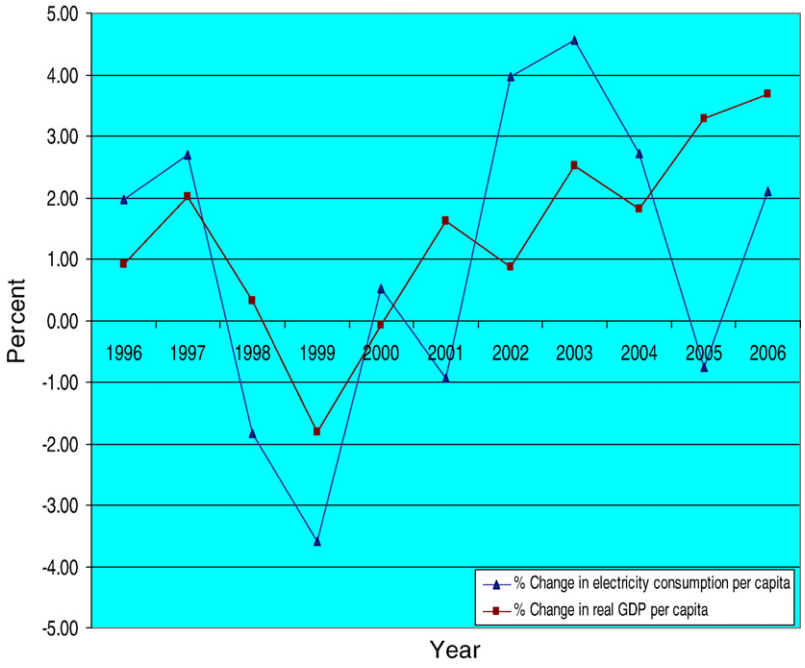


Fig. 1. Trends of energy consumption and economic growth during the period 1996–2006. Source: author's own computations from the World Development Indicators CD-ROM, 2007; International Energy Agency, 2008; IFS Yearbook, 2007.

variable in a bivariate framework—thereby creating a simple trivariate causality model.

The rest of the paper is structured as follows: Section 2 discusses the trends of electricity consumption and economic growth in South Africa. Section 3 presents the literature review, while Section 4 deals with the empirical model specification, estimation technique and the empirical analysis of the regression results. Section 5 concludes the study.

2. Electricity consumption and economic growth in South Africa

South Africa is considered to be the highest electricity producer and consumer in Africa. It is estimated that more than half of the electricity generated in Africa comes from South Africa. Likewise, the country is historically considered to have one of the highest electricity reserve margins. However, the reserve margin has recently decreased somewhat. For example, the electricity reserve margin in South Africa decreased from 25% in 2002 to 20% in 2004, and there after to 16% in 2006. The reserve margin estimate for 2008 is projected to be between 8–10%, which is far lower than the target set for the country, i.e. a minimum of 15%. The South African government attributes the unprecedented decline in the electricity reserve margin during the recent years to the robust economic growth and the associated demand for electricity. Indeed the electricity consumption, especially from the residential sector has increased phenomenally since 1994. The electric power consumption/capita, for example, increased from 3644.44 kWh in 1980 to 4433.59 kWh in 1995, and later to 4521.17 kWh in 1996 and 4642.91 kWh in 1997. Although the consumption decreased somewhat between 1997 and 1999, the consumption later increased to 4884.79 kWh in 2004. Currently, the electricity consumption/capita is estimated at about 5039.7 kWh.

It is also worth noting that South Africa has one of the lowest electricity prices in the world. Even after the recent increase in the electricity price, electricity in South Africa is still one of the cheapest in the world. The continued demand for electricity in South Africa and the need to restore a workable reserve margin forced Eskom, a parastal responsible for the supply of electricity in South Africa, to embark on a number of interventions in early 2008, which included a national awareness campaign, among others. In some instances, load shedding had to be used as the last resort in order to prevent a system-wide blackout. These interventions enabled Eskom to slightly bring the demand for electricity closer to its supply, while at the same time maintaining a reasonable reserve margin. Currently, the government is working on both medium and long-term programmes, which are meant to enable the country to cope with the future demand for electricity. At the same time, alternative sources of energy are being explored, in order to supplement the supply of electricity. Fig. 1 shows the trends of electricity consumption/capita vis-à-vis the trends of real GDP/capita during the period 1996–2006.

3. Literature review

The relationship between energy consumption and economic growth has been examined extensively in the literature since the seminal work of Kraft and Kraft (1978), but with conflicting results. Kraft and Kraft (1978), for example, while conducting a causality test between energy consumption and GNP for the USA, found evidence of a unidirectional causality running from GNP to energy consumption. However, studies conducted by Akarca and Long (1979, 1980) on the causal relationship between energy consumption and GNP for the USA, failed to confirm Kraft and Kraft's (1978) findings. Currently, four views exist in the literature regarding the relationship between energy consumption and economic growth (see Odhiambo, 2009). The first view argues that energy consumption Granger-causes economic growth. The second view argues that it is economic growth that drives the consumption of energy in many countries, and that as the economy grows the demand for energy from different sections of the economy automatically increases. The third view, however, maintains that both energy consumption and economic growth Granger-cause each other, i.e., that there is a bidirectional causality between energy consumption and economic growth. Finally, the fourth view maintains that there is no causality between energy consumption and economic growth. In other words, these studies assert that energy consumption and economic growth are neutral with respect to each other.

The empirical work that is consistent with the first view includes Yu and Choi (1985) for the case of The Philippines, Cheng (1997) for the case of Brazil, Chang et al. (2001) for the case of Taiwan, Soytas and

Sari (2003) for the case of Turkey, France, Germany and Japan, Shiu and Lam (2004) for the case of China, Wolde-Rufael (2004) for the case of Shanghai, Lee (2005) for the case of 18 developing countries, Altinay and Karagol (2005) for the case of Turkey, Wolde-Rufael (2006) for the case of Benin, the Democratic Republic of Congo and Tunisia and Narayan and Singh (2007) for the case of Fiji, Odhiambo (2009) for the case of Tanzania, among others. Chang et al. (2001), for example, while examining the temporal causality between energy consumption, employment and output, found a distinct unidirectional causality running from energy consumption to economic growth in Taiwan. The authors concluded that energy conserv ation will restrain the output growth in Taiwan. Likewise, Lee (2005), while examining the relationship between energy consumption and GDP in developing countries, found that energy consumption Granger-causes economic growth in the study countries. More recently, Narayan and Singh (2007) investigated the nexus between electricity consumption and economic growth in Fiji. Using a multivariate framework, the authors found a distinct unidirectional causality running from electricity consumption to economic growth. The authors concluded that since Fiji is an energy-dependent country, energy conservation policies are likely to have adverse effects on its economic growth.

Contrary to the above studies, there are a number of studies that contend that it is economic that Granger-causes energy consumption. Ghosh (2002), while conducting the causal relationship between electricity consumption and economic growth in India, found a distinct unidirectional causality running from economic growth to electricity consumption without any feedback effect. The author, therefore, concluded that electricity policies can be initiated without any deteriorating economic effects (Ghosh, 2002: 125). Wolde-Rufael (2006), while examining the causal relationship between electricity consumption and economic growth in 17 African countries, found a positive unidirectional causality running from real GDP/capita to electricity consumption/capita in six countries, namely Cameroon, Ghana, Nigeria, Senegal, Zambia and Zimbabwe. Narayan and Smyth (2005), while examining the relationship between electricity consumption, employment and real income in Australia, found that real income Granger-causes electricity consumption in Australia. Other studies that have found a unidirectional causality running from economic growth to electricity consumption include Fatai et al. (2004) for the case of Australia, and Thoma (2004) for the case of the USA, among others.

Despite the above studies, there are a number of studies that have found evidence of a bidirectional causality between energy consumption and economic growth. Masih and Masih (1997), for example, while examining the causal relationship between energy consumption and economic growth, found a bidirectional causality between energy consumption and real income in Korea and Taiwan. Likewise, Morimoto and Hope (2004) found a bidirectional causality between electricity consumption and economic growth in Sri Lanka during the period 1960–1998. Yang (2000), while investigating the causal relationship between energy and GDP using updated Taiwan data for the period 1954–1997, found a bidirectional causality between total energy consumption and GDP. While examining the causal relationship between energy consumption and economic growth in India, Paul and Bhattachrya (2004) found a bidirectional causality to exist between energy consumption and economic growth in India. In the same vein, Jumbe (2004), while examining the relationship between electricity consumption and economic growth in Malawi, found a two-way causal relationship between electricity consumption and economic growth for the period 1970–1999. Other studies that have found evidence of a bidirectional causality between energy and economic growth include Soytas and Sari (2003) for the case of Argentina, Wolde-Rufael (2006) for the case of Egypt, Gabon and Morocco, and Glosure and Lee (1997) for the case of South Korea and Singapore, among others.

Although the majority of the previous studies have found a causal relationship to exist between energy consumption and economic growth, the empirical evidence of some previous studies have shown that no causality exists between energy and economic growth. In other words, these studies maintain that energy and economic growth are neutral with respect to each other. These include Yu and Hwang (1984) for the case of the USA, Yu and Choi (1985) for the case of the USA, the United Kingdom and Poland, and Altinay and Karagol (2004) for the case of Turkey, among others.

4. Estimation techniques and empirical analysis

4.1. Empirical model specification

In this section, a dynamic Granger causality test is used to examine the direction of causality between electricity consumption and economic growth in South Africa. The Granger causality method is chosen in this paper over other alternative techniques because of its favourable response to both large and small samples (see Odhiambo, 2008). Unfortunately, causality studies based on a bivariate framework have been found to be very unreliable as the introduction of a third important variable can change the both the inference and the magnitude of the estimates (see also Caporale and Pittis, 1997; Caporale et al., 2004; Odhiambo, 2008). Given the weakness associated with the bivariate causality framework, the current study examines the causal relationship between electricity consumption and economic growth by incorporating employment as an intermittent variable—thereby creating a simple trivariate model. The trivariate Granger causality model between electricity consumption, employment and economic growth based on the error–correction mechanism can be expressed as follows:

$$\mathrm{lny}/N_t = \lambda_0 + \sum_{i=1}^{m} \lambda_{1i}\mathrm{lny}/N_{t-i} + \sum_{i=1}^{n} \lambda_{2i}\mathrm{lnELEC}_{t-i} + \sum_{i=1}^{n} \lambda_{3i}\mathrm{lnEMP}_{t-i} + \lambda_4 \mathrm{ECM}_{t-1} + \mu_t \quad (1)$$

$$\mathrm{lnELEC}_t = \varphi_0 + \sum_{i=1}^{m} \varphi_{1i}\mathrm{lny}/N_{t-i} + \sum_{i=1}^{n} \varphi_{2i}\mathrm{lnELEC}_{t-i} + \sum_{i=1}^{n} \varphi_{3i}\mathrm{lnEMP}_{t-i} + \varphi_4 \mathrm{ECM}_{t-1} + \varepsilon_t \quad (2)$$

$$\mathrm{lnEMP}_t = \delta_0 + \sum_{i=1}^{m} \delta_{1i}\mathrm{lny}/N_{t-i} + \sum_{i=1}^{n} \delta_{2i}\mathrm{lnELEC}_{t-i} + \sum_{i=1}^{n} \delta_{3i}\mathrm{lnEMP}_{t-i} + \delta_4 \mathrm{ECM}_{t-1} + \nu_t \quad (3)$$

Where: ln y/N is the log of real/capita income; lnELEC is the log of electricity consumption/capita; ln EMP is the log of employment rate; ECM $_{t-1}$ is the error correction term lagged one period; and μ, ε and ν_t are mutually uncorrelated white noise residuals.

In addition to indicating the direction of causality amongst variables, the error–correction mechanism also enables us to distinguish between the short-run and the long-run Granger causality. The "short-run" causal impact is measured through the F-statistics and the significance of the lagged changes in the independent variables, whereas the "long-run" causal impact is measured through the significance of the t-test of the lagged error–correction term. Based on the Eqs. (1)–(3), the following causal relationships can be derived between electricity consumption, employment and economic growth.

Causal flow	Conditions
(1) Electricity consumption (ELEC) → economic growth (y/N)	$\lambda_{2i} \neq 0$; $\lambda_4 \neq 0$
(2) Economic growth (y/N) → electricity consumption (ELEC)	$\varphi_{1i} \neq 0$; $\varphi_4 \neq 0$
(3) Employment (EMP) → electricity consumption (ELEC)	$\varphi_{3i} \neq 0$; $\varphi_4 \neq 0$
(4) Electricity consumption (ELEC) → employment (EMP)	$\delta_{2i} \neq 0$; $\delta_4 \neq 0$
(5) Economic growth (y/N) → employment (EMP)	$\delta_{1i} \neq 0$; $\delta_4 \neq 0$
(6) Employment (EMP) → economic growth (y/N)	$\lambda_{3i} \neq 0$; $\lambda_4 \neq 0$

Table 1
Stationarity test of variables in levels—DF-GLS test.

Variable	No trend	Trend
Ly/N	−1.455959	−1.512738
LELEC	0.029443	−2.494920
LEMP	−1.330957	−1.635293

Note: Critical values are based on Elliot et al. (1996), Table 1.

Table 2
Stationarity test of variables in levels—Philips–Perron (PP) test.

Variable	No trend	Trend
Ly/N	−0.938185	−0.110233
LELEC	−1.811589	−2.727949
LEMP	−2.350701	−2.452025

Note: The truncation lag is based on Newey and West (1987) bandwidth.

4.2. Data sources and definitions of variables

Annual time series data, which covers the period 1971 to 2006 period, are utilised in this study. All the variables are expressed in logarithmic form. The data sources and definitions of the variables are highlighted below:

i) Electricity consumption

The residential electricity consumption/capita is calculated as: residential electricity consumption/capita = Total residential electricity/Total population. The data was obtained from the International Energy Agency.

ii) Real/capita GDP

The real/capita GDP data are obtained from the South African Reserve Bank Quarterly Bulletin.

iii) Employment level

The employment level in this study is proxied by an index of manufacturing sector employment (see also Narayan and Smyth, 2005). The data was obtained from the IMF International Financial Statistics Yearbook, 2007.

4.3. Empirical analysis

4.3.1. Stationarity test

Before the variables are differenced, the stationarity tests on all variables used in the causality test are performed. The results of the unit root tests are presented in Tables 1 and 2.

The results of the stationarity tests reported in Tables 1 and 2 show that all the three variables are non-stationary in levels. Having found that the variables are not stationary in levels, the next step is to difference the variables once in order to perform stationarity tests in first differences. The results of the stationarity tests in first differences–based on the DF-GLS and Ng-Perron tests–are reported in Tables 3 and 4.

The results reported in Tables 3 and 4 show that after differencing the variables once, all the variables were confirmed to be stationary.

Table 3
Stationarity test of variables in first differences—DF-GLS test.

Variable	No trend	Trend
DLy/N	−3.3630***	3.7224**
DLELEC	−5.5880***	−5.7349***
DLEMP	−3.3279***	−4.34502***

Note: 1) Critical values are based on Elliot et al. (1996), Table 1.
2) ** and *** denote significance at 5% and 1% respectively.

Table 4
Stationarity test of variables in first differences—Philips–Perron (PP) test.

Variable	No trend	Trend
DLy/N	−4.0144***	−4.10147**
DLELEC	−5.5785***	5.5567***
DLEMP	−4.1720***	4.26848***

Note: 1) The truncation lag is based on Newey and West (1987) bandwidth.
2) ** and *** denote significance at 5% and 1% respectively.

The DF-GLS and Phillips–Perron tests applied to the first difference of the data series reject the null hypothesis of non-stationarity for all the variables used in this study. It is, therefore, concluded that all the variables used in this study are integrated of order one.

4.3.2. Cointegration test

Having confirmed that all variables included in the causality test are integrated of order one, the next step is to test for the existence of a cointegration relationship between electricity consumption (ELEC), employment (EMP) and economic growth (y/N). For this purpose, the study uses the Johansen–Juselius (maximum likelihood) cointegration

Table 5
Maximum likelihood cointegration test.

Trace test				Maximum Eigen value test			
Null	Alternative	Statistics	95% Critical value	Null	Alternative	Statistics	95% Critical value
Cointegration between Ly/N, LELEC/N and LEMP							
$r=0$	$r\geq1$	36.72	29.7	$r=0$	$r=1$	28.35	21.0
$r\leq1$	$r\geq2$	8.365	15.4	$r\leq1$	$r=2$	6.352	14.1
$r\leq2$	$r\geq3$	2.013	3.8	$r\leq2$	$r=3$	2.013	3.8

Notes:
1) r stands for the number of cointegrating vectors.
2) The lag structure of VAR is determined by the highest values of the Akaike information criterion and Schwartz Bayesian Criterion.

Table 6
Causality tests between ΔLy/N, ΔLELEC and ΔLEMP.

Variables in equation	Equation 1 dependent variable (ΔLy/N)	Equation 2 dependent variable (ΔLELEC)	Equation 3 dependent variable (ΔLEMP)
Constant	−0.00241(−0.578)	0.03077(3.472)	0.0034358(0.412)
ΔLy/N	–	0.88875(3.086)***	0.27476(0.799)
ΔLy/N-1	0.28580(1.523)	0.56825(1.489)	0.14561(0.407)
ΔLy/N-3	0.42977(2.08)**	–	–
ΔLELEC	0.23914(2.563)**	–	–
ΔLELEC-1	0.15053(1.329)	−0.17364(−0.981)	−0.26228(−1.492)
ΔLELEC-2	–	−0.34461(−2.082)**	0.082613(0.491)
ΔLELEC-4	–	−0.25873(−1.549)	–
ΔLEMP	0.28571(2.205)**	–	–
ΔLEMP-1	0.074032(0.493)	−0.34864(−1.241)	0.27466(0.894)
ΔLEMP-2	–	0.23435(0.23435)	0.25672(0.890)
ΔLEMP-3	−0.17328(−1.144)	–	–
ΔLEMP-4	–	–	0.27456(1.043)
ECM $_{-1}$	−0.20629(−2.072)**	−0.19367(−2.766)***	−0.14247(−0.845)
F-test	3.3572 [0.0133]	2.8873 [0.0259]	0.91737 [0.5226]
R^2	3.3572 [0.0133]	0.535	0.26844
DW	2.17	2.04	1.80
Diagnostic tests			
AR 1-2F (2, 18)	1.0912 [0.3570]	0.39082 [0.6821]	0.18798 [0.8302]
ARCH 1 F (1, 18)	0.8228 [0.3763]	0.35003 [0.5615]	2.0722 [0.1672]
Normality X^2	2.7925 [0.2475]	4.0119 [0.1345]	0.21508 [0.9840]
RESET	0.1023 [0.7526]	3.8454 [0.0647]	1.3997 [0.2514]

Notes: *, ** and *** denote 1%, 5% and 10% level of significance respectively. The numbers in parentheses represent t-statistics.

Table 7
Summary of causality tests

Variables	Causality	General conclusion
ΔLy/N (dependent variable), ΔLELEC and ΔLEMP	There is a causal flow from both electricity consumption and employment to economic growth.	Both electricity consumption and employment Granger-cause economic growth.
ΔLELEC (dependent variable), ΔLy/N and ΔEMP	There is a causal flow from economic growth to electricity consumption.	Economic growth Granger-cause electricity consumption
ΔLEMP (dependent variable), ΔLy/N and ΔLELEC	There is no causal flow from either electricity consumption or economic growth to employment.	Neither electricity consumption nor economic growth Granger-causes employment.

test procedure. The results of the Johansen–Juselius cointegration tests are presented in Table 5.

The results of Johansen–Juselius cointegration tests reported in Table 5 indicate the existence of a stable long-run relationship between electricity consumption (ELEC), employment (EMP) and economic growth (y/N). Both the trace test and the maximum Eigen value statistics reject the null hypothesis of no cointegration. Specifically, the results show that there is a unique cointegrating vector between electricity consumption, employment and economic growth.

4.3.3. Analysis of causality test based on error–correction model

The results of the general (over-parameterised) causality tests between electricity consumption, employment and economic growth (not presented here) are difficult to interpret and many variables are not significant—as expected. The electricity consumption (ELEC), employment (EMP) and economic growth (y/N) equations are, therefore, reduced until the preferred models are obtained. A summary of the results of the preferred models of the causality test between electricity consumption (ELEC), employment (EMP) and economic growth (y/N) is displayed in Table 6.

The empirical results reported in Table 6 show that there is a bidirectional causality between electricity consumption and economic growth. The long-run causality from electricity consumption to economic growth is supported by the coefficient of the lagged error–correction term, which is negative and statistically significant, as expected. The short-run causality from electricity consumption to economic growth is, however, supported by the *F*-test and the coefficient of the electricity consumption variable in the economic growth function, which are both statistically significant. The reverse causality from economic growth to electricity consumption, on the other hand, is supported by the lagged error–correction term, the *F*-statistic and the coefficient of the economic growth variable in the electricity consumption function—which are all statistically significant. In addition, the results show that there is a unidirectional causality from employment to economic growth. This finding is supported by the coefficient of the employment variable in the economic growth, which is positive and statistically significant—as expected. The results, however, failed to find any direct causal relationship between electricity consumption and employment. The results apply irrespective of whether the causality is estimated in the short-run or in the long-run dynamic. A summary of the causality test between the three variables is presented in Table 7.

5. Conclusion

This study examines the causal relationship between electricity consumption and economic growth using the South African data. The study attempts to answer one critical question. Is there a causal relationship between electricity consumption and economic growth in South Africa? Previous studies on this subject suffer from two major limitations. Firstly, the majority of the studies has mainly used a bivariate causality model, and may, therefore, suffer from the omission-of-variables bias. Secondly, some of the studies have over-relied on the cross-sectional data, which may satisfactorily address the country-specific issues. Unlike the majority of the previous studies, the current study incorporates the employment rate as an intermittent variable in the bivariate model—thereby creating a simple trivariate causality framework. Using the cointegration-based error–correction mechanism, the empirical results of this study show that there is a bidirectional causality between electricity consumption and economic growth in South Africa. In addition, the study finds that there is a distinct unidirectional causality from employment to economic growth. The results apply irrespective of whether the causality is estimated in the short-run or in the long-run dynamics. The study, therefore, recommends that policies geared towards the expansion of the electricity infrastructure should be intensified in South Africa in order to cope with the increasing demand exerted by the country's strong economic growth and rapid industrialisation programme. This will certainly enable the country to avoid unprecedented power outages similar to those experienced in the country in mid-January 2008.

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